



Technical note

Transversal dose profile reconstruction for clinical proton beams: A detectors inter-comparison



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ABSTRACT

Purpose: The main purpose of this work is the inter-comparison between different devices devoted to the transversal dose profile reconstruction for daily QA tests in proton therapy.

Methods: The results obtained with the EBT3 radiochromic films, used as a reference, and other common quality control devices, have been compared with those obtained with a beam profiling system developed at the “Laboratori Nazionali del Sud” of Italian Institute for Nuclear Physics (INFN-LNS, Catania, Italy). It consists of a plastic scintillator screen (thickness 1 mm), mounted perpendicularly to the beam axis and coupled with a highly sensitive CCD detector in a light-tight box.

Results and conclusion: The tests, carried out both at the INFN-LNS and Trento Proton Therapy Center facilities, show, in general, a good agreement between the different detectors. The beam profiling system, in particular, appears to be a promising quality control device for 2-D relative dosimetry, because of its linear response in a dose rate range useful for proton therapy treatments, its high spatial resolution and its short acquisition and processing time.

1. Introduction

Hadrontherapy represents the most advanced form of external radiation modality in tumor treatments. It makes use of high energetic (60 AMeV–480 AMeV) proton/ion beams accelerated by cyclotrons or synchrotrons and exploits the increased selectivity of charged particles in terms of dose released and biological effectiveness compared to photons [1–3].

Nowadays (2019), more than 80 facilities are active and the number is constantly rising in recent years, thus demonstrating the growing interest in the clinical use of charged particles for cancer treatment [4]. In Italy three particle therapy facilities are in operation: the Trento Proton Therapy facility [5], that started clinical operations in October 2014 and where protons are accelerated by a normal-temperature cyclotron (Proteus-235, IBA Company); the Centro Nazionale di Adroterapia Oncologica (CNAO), active since 2011, where proton and carbon beams with energies up to 250 MeV and 400 AMeV respectively, are accelerated by a PIMMS/TERA synchrotron [6,7]; and the CATANA (Centro di AdroTerapia e Applicazioni Nucleari Avanzate) facility

[8–11], active since 2002 at the Laboratori Nazionali del Sud – Istituto Nazionale di Fisica Nucleare (INFN-LNS) in Catania (I), where 62 MeV protons, accelerated by a superconducting cyclotron, are used for the radio-therapeutic treatment of ocular diseases.

Safe and accurate treatments are guaranteed by daily Quality Assurance (QA) tests that are performed in any facility. The excellent results obtained so far, in terms of treatment accuracy are, in fact, reached thanks to the frequent, reproducible and precise dosimetric checks on both transverse and depth dose proton beams distributions. Among these tests, the periodic verification of beam homogeneity is of importance in any QA program. All checks are carried out using several kinds of radiation detectors as ionization chambers [12], silicon diodes and photodiodes [13], radiographic and radiochromic films [14]. In particular, radiochromic films are widely used in daily QA as reference detectors for the homogeneity determination; nevertheless, despite the advantage of their high spatial resolution, response homogeneity and stability, they are quite expensive, time consuming and do not provide online results. Therefore, alternative detectors have been proposed in the last years and are now widely used worldwide [15–18].

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In this paper, we report the results of an inter-comparison between different devices devoted to the transversal dose profile reconstruction, carried out both at the CATANA and Trento Proton Therapy facilities. The performance of a scintillator-based beam profiling (BP) system entirely developed at the INFN-LNS, in particular, were compared with other quality control devices, routinely adopted in QA tests. The preliminary characterization of the BP system in terms of short-term stability and dose-rate dependence has been performed as well.

2. Material and methods

2.1. Experimental setup

The inter-comparison measurements were performed with two different proton beam energies (62 and 100 MeV), using EBT3 radiochromic films [19] as reference detectors. The BP system was firstly characterized and studied at the INFN-LNS proton therapy beamline, where the QA checks on the lateral dose are performed before any patient treatment by using a p-type silicon diode detector [21] and EBT3 films. A detailed description of the clinical beamline, of its main elements and dosimetric devices and procedures, can be found elsewhere [8–11]. The BP system response was successively compared with the commercial Lynx (IBA Company) [22], the silicon pixelated Timepix detector (Medipix collaboration) [23] and EBT3 at the TIFPA-INFN proton research beamline inside the Trento Proton Therapy facility [5,25], where the available cyclotron accelerates proton beam in the 70–228 MeV energy range, with currents spanning between 1 and 320 nA [5,26].

All irradiations were performed at the beam entrance by positioning the surface of the detectors at the isocenter plane, perpendicularly to beam direction, 3 m and 1.25 m behind the beam exit window of CATANA and TIFPA beamlines, respectively. No under-response or quenching effects are expected in this low LET region for all the above detectors. The passive scattering technique was used to spread the proton beam ensuring a uniform dose profile, while downstream brass collimators enable the final beam collimation; beam spot was circular, 30 mm in diameter, at the CATANA facility, and rectangular, 10 mm × 12 mm, at the higher energy TIFPA beamline.

2.2. Scintillator-based beam profiling system

The system described in this paper is an improved version of that already developed at INFN-LNS in 2002 [27,28].

This updated version, specifically designed to improve the performances of the system in terms of signal-to-noise ratio, spatial resolution, acquisition and processing time and analysis algorithm, consists of a 50 mm × 50 mm EJ-204 (Eljen Technology, Sweetwater, TX, USA) plastic scintillator screen (thickness of 1 mm), characterized by high scintillation efficiency, high speed and good attenuation length (see Table 1), mounted perpendicularly to the beam axis at a fixed distance from the beam exit. The EJ-204 scintillator consists of a polyvinyltoluene (~97%) matrix doped with organic fluors; it is almost water equivalent in terms of electron density and it is suitable for applications in relative dosimetry for its high conversion efficiency in

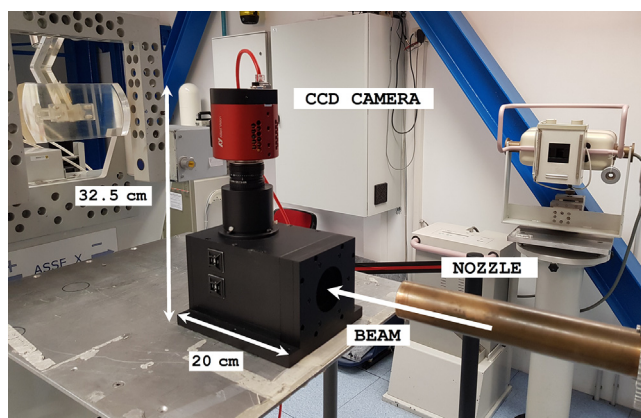


Fig. 1. The beam monitoring system ready for the acquisition in the CATANA treatment room.

light emission [29].

The light emitted when the scintillator is hit by the proton beam is detected by an highly sensitive, Peltier cooled, low-noise 14-bit monochromatic CCD camera (Bigeye G-283 Cool Allied Vision Technologies GmbH, Germany) with a resolution of 1928 × 1452 pixels (4.54 μm wide square) coupled with an high optical imaging performance Xenon 0.95/25 lens (Schneider Kreuznach GmbH, Germany) and processed by an in-house software. A mirror, positioned at 45°, behind the scintillator at a distance of 80 mm, reflects the light coming from the scintillator itself thus avoiding the direct irradiation of the CCD-chip. All the components of the system are assembled in Teflon light-tight box that can be easily transported. The system case coupled with the CCD camera and ready for the acquisition, is shown in Fig. 1 mounted at the irradiation point of the CATANA protontherapy beamline.

The CCD camera is firmly fixed to the box by means of adjustment screws permitting its accurate alignment; it features an outstanding signal-to-noise ratio and a sensitivity of 73% at 530 nm, being the maximum quantum efficiency for wavelengths from 500 nm to 650 nm [30]. The signal is acquired and stored as a 14-bit 1928 × 1452 pixels matrix of integer.

2.2.1. Software analysis and derived dosimetric parameters

The real-time data acquisition, the quantitative analysis of the beam profiles and the calculation of the dosimetric-relevant parameters along both horizontal (X) and vertical (Y) directions, were entirely performed using specific in-house software libraries, developed on the LabView platform (National Instruments, Austin, TX, USA).

Firstly, a simple algorithm which calculates the pixel size in real dimensions (calibration factor) has been implemented; for the adopted optical distance, we measured a calibration factor of 0.045 mm/pixel, that means that 1 pixel on the matrix covers a real area of 2.025 μm^2 .

Once calibration is carried out, the application performs an arithmetical mean of a predefined number of consecutive acquired frames and subtracts the background, i.e the image acquired under the same conditions but without irradiation. Transverse orthogonal dose profiles are then extracted by the final image as distributions of the pixels gray values. Each acquired profile is analyzed and a set of well-defined, clinically accepted parameters (see Table 2), connected to the beam quality (radiation field, field ratio, lateral penumbras [10,27], flatness and symmetry [31]) are calculated.

All measurements reported in this work were performed following the same experimental procedure: the time for the acquisition (exposure time) of a single 1928 × 1452 pixels frame was fixed at 20 ms, while the final images were obtained averaging over 1000 CCD frames in order to reduce the noise, both thermal and electronic, and minimizing the image associated uncertainty. Maximum total acquisition

Table 1

Physical characteristics of EJ-204 plastic scintillator.

Light output, [% anthracene]	68
Scintillation Efficiency, [photons/1 MeV e^-]	10,400
Wavelength of max. emission, [nm]	408
Rise time, [ns]	0.7
Decay time, [ns]	1.8
Pulse width, FWHM, [ns]	2.2
Refractive index	1.58
No. of electrons per cm^3 [$\times 10^{23}$]	3.33
Density, [g/cc]	1.023

Table 2
Beam quality parameters.

RADIATION FIELD	Full Width at Half Maximum (FWHM) of the profile; it allows evaluating the size of the proton beam
FIELD RATIO	The ratio of the profile widths at 90% and 50% of the maximum dose; it quantifies the difference between the experimental profile and an ideal rectangular shape
LATERAL PENUMBRAS	Lateral distance between the 80% and 20% of maximum dose points on both sides of the profile; it gives the value of right and left slopes of lateral dose profiles
FLATNESS	Maximum permissible percentage variation from the average dose across the central 80% of the FWHM of the profile in a plane transverse to the beam axis: $Flatness = \left(\frac{D_{max} - D_{min}}{D_{max} + D_{min}} \right) \times 100\% \quad (1)$ where D_{max} and D_{min} are the maximum and minimum dose values in the central 80% of the profile respectively
SYMMETRY	Maximum permissible percentage deviation of the “left-side” dose from the “right-side” dose of a beam profile often at 80% of the FWHM points: $Symmetry = \left(\frac{D(x)}{D(-x)} \right) \times 100\% \quad (2)$ where $D(x)$ and $D(-x)$ are symmetrical points with respect to the beam axis

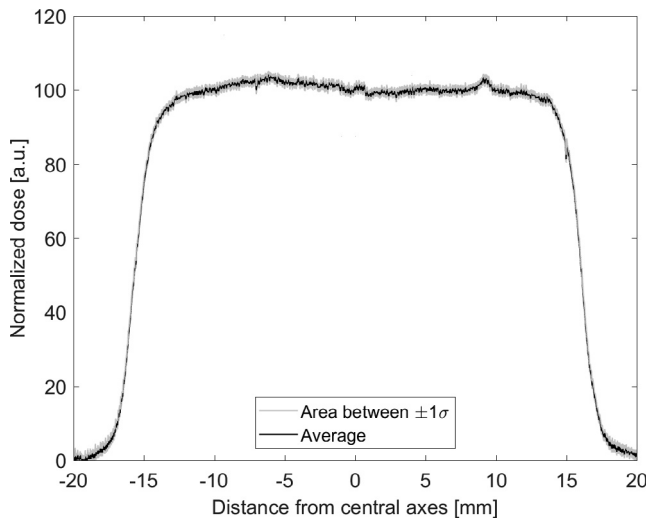


Fig. 2. Averaged experimental dose distribution along the horizontal direction, evaluated on the basis of eight consecutive irradiations under the same proton beam conditions. Data are normalized to central axes; errors (gray area) are estimated as the standard deviation ($\pm 1\sigma$). Similar results are obtained for the Y profile.

Table 3
Average and variance of the relevant beam parameters averaged for the eight acquisitions.

	PROFILE X	PROFILE Y
FWHM [mm]	$31.737 \pm 0.2\%$	$31.901 \pm 0.1\%$
90/50 Ratio	$0.901 \pm 0.8\%$	$0.873 \pm 0.9\%$
Right penumbra [mm]	$1.548 \pm 2.0\%$	$1.951 \pm 1.7\%$
Left penumbra [mm]	$1.678 \pm 2.3\%$	$2.028 \pm 2.5\%$
Symmetry [%]	104.889 ± 1.570	103.623 ± 1.970

time, in these conditions, was 20 s. The measurements were performed with the maximum aperture of the CCD iris to ensure the maximum efficiency in collecting the light produced by the scintillating screen. A region of interest (ROI), entirely contained inside the beam spot, was then extracted from the averaged image and used for the analysis. The ROI extension has been chosen in order to be small enough to result homogeneous and sufficiently large to be statistically significant.

A correction procedure, useful to correct for any purely optical effect (e.g. vignetting or image distortion), has not been implemented yet, while a smoothing algorithm to avoid spikes, which can arise from the interaction between the secondary ionizing radiation produced in the system components and the CCD-chip or from isolated broken pixel, can be applied on the 1-D transversal dose profiles; non-completely removed spikes are identified by the analysis software and not taken into

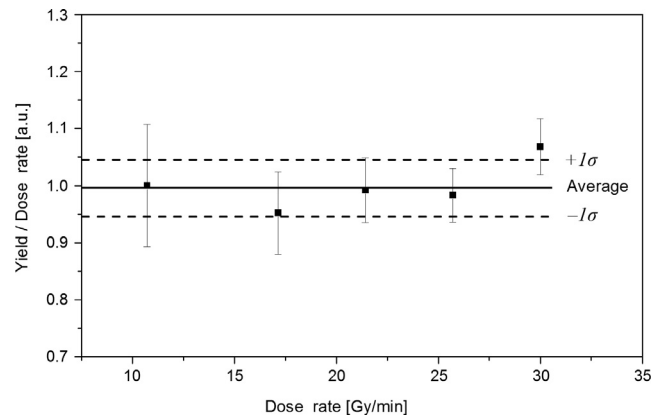


Fig. 3. Normalized system response as a function of the dose rate for 62 MeV proton beam. Yield corresponds to the mean pixel value of one thousand acquired frames in the selected ROI. Error bars are estimated as the standard deviation; 1σ lines represent the standard error of the mean of the five values.

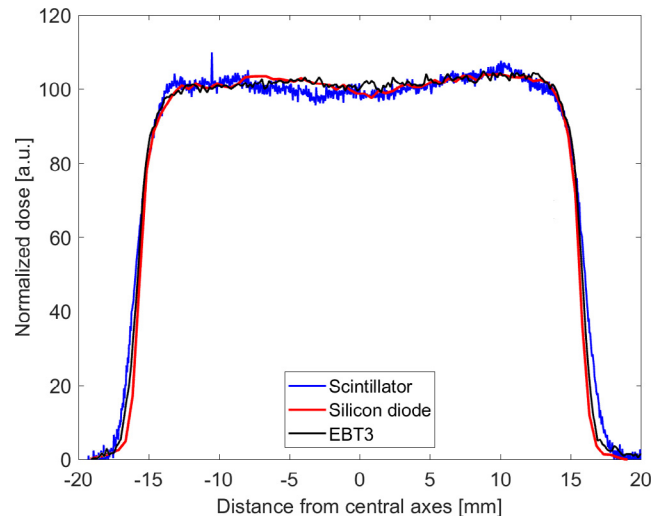


Fig. 4. Comparison between transversal dose profiles measured with the BP system, the silicon diode and the EBT3 for 62 MeV protons along the horizontal direction. Similar results are obtained for the vertical profile.

account in the calculation of the beam quality parameters.

2.2.2. System response characterization

The preliminary characterization of the BP system was carried out at the CATANA proton therapy facility where short-term stability, i.e. stability into reconstruct sharp dose profiles when repeating measurements over time under the same experimental conditions, and linearity

Table 4

Beam quality parameters extracted for the scintillator-based system, the silicon diode and the EBT3 for 62 MeV protons.

PROFILE X	EBT3	EJ-204	DIODE
FWHM [mm]	31.665 ± 0.450	31.970 ± 0.305	31.655 ± 0.315
90/50 Ratio	0.930 ± 0.029	0.921 ± 0.014	0.937 ± 0.019
Right penumbra [mm]	1.185 ± 0.115	1.530 ± 0.105	1.210 ± 0.095
Left penumbra [mm]	1.220 ± 0.125	1.575 ± 0.110	1.250 ± 0.090
Flatness [%]	3.580 ± 0.315	3.220 ± 0.160	3.040 ± 0.240
Symmetry [%]	105.430 ± 1.580	106.502 ± 1.595	105.512 ± 1.055

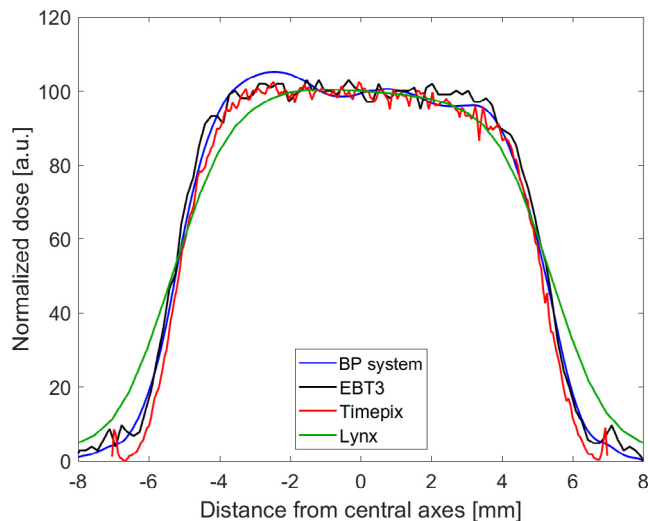


Fig. 5. Comparison between transversal dose profiles measured with the BP system, the Lynx detector, the Timepix and the EBT3 for 100 MeV protons along the horizontal direction. Similar results were obtained for the vertical profile. No smoothing algorithm was applied in the profiles from EBT3 and Timepix.

with the beam dose rate were verified.

The short-term stability, calculated as the percentage ratio of the standard deviation to the mean CCD output, was evaluated performing eight consecutive irradiations with identical dose-rate values and resulted to be smaller than 0.9%, indicating a good performance of the system. The resulting X averaged experimental transversal dose distribution, normalized to central axes, is reported in Fig. 2, together with the experimental errors calculated as the standard deviation.

For each curve, the relevant clinical parameters (Section 2.2.1) were derived as well, and their average and variance evaluated. Results are reported in Table 3, showing uncertainties within 2.5% for each of the studied parameters.

The system response as a function of the absorbed dose rate was evaluated in the range between 10.5 Gy/min and 30.0 Gy/min of the monochromatic 62 MeV proton beam, corresponding to 9.0×10^9 and 2.7×10^{10} pps/cm², respectively. Fig. 3 reports the ratio between the scintillating yield signal, calculated as the average pixel value in the ROI and normalized to the minimum dose rate value (10.5 Gy/min), with respect to the dose rate.

Table 5

Beam quality parameters extracted for the scintillator-based system, the Lynx and Timepix detectors and the EBT3 for 100 MeV protons.

PROFILE X	EBT3	EJ-204	TIMEPIX	LYNX-IBA
FWHM [mm]	10.460 ± 0.330	10.535 ± 0.100	10.450 ± 0.120	10.750 ± 0.500
90/50 Ratio	0.793 ± 0.035	0.768 ± 0.013	0.765 ± 0.012	0.656 ± 0.074
Right penumbra [mm]	1.370 ± 0.115	1.400 ± 0.095	1.385 ± 0.105	2.350 ± 0.550
Left penumbra [mm]	1.220 ± 0.120	1.295 ± 0.090	1.260 ± 0.100	2.350 ± 0.550
Flatness [%]	4.125 ± 0.290	4.314 ± 0.215	3.695 ± 0.165	0.825 ± 0.350
Symmetry [%]	101.978 ± 1.580	103.003 ± 1.545	102.864 ± 1.543	101.343 ± 1.520

The plot shows a constant ratio within the experimental errors, confirming a remarkable linear behavior of the BP over a dose rate range typically adopted in the proton therapy clinic. The signal increases linearly as a function of the dose rate with a correlation fitting factor $R^2 > 0.99$. No saturation effects are observed, being the light output at the maximum delivered dose rate well below the saturation limit of the system.

No significant image geometrical distortions between different dose rates are registered over the observed dynamic range for the normalized beam profiles along both the X and Y directions. The image width from the scintillator does not increase with particle intensity, indicating an excellent stability of the screen, and no broadening effects occur throughout the detector with increasing dose rate.

2.3. Detectors inter-comparison

The results obtained with the BP system have been compared with those from other routinely-used devices (see § 3.1 and 3.2). The detectors that have been considered are:

–EBT3 radiochromic films (International Specialty Products, Wayne, NJ, USA) that are considered a reference in relative dose distribution reconstruction [19,20]. After receiving a given dose the films were scanned using a 48-bit RGB Epson Expression 10000XL commercial flatbed scanner in transmission mode; the intrinsic spatial resolution was set to 0.17 mm/pixel, corresponding to 150 dpi acquisition, the 16-bit depth values were extracted in the red channel from the RGB image by means of the ImageJ (v1.51) freeware software, converted in gray scale and stored in TIFF format. The analysis was carried out using a dedicated routine in which a correction, which takes into account the calibration curve (optical density vs. dose), was implemented.

–A motorized p-type silicon diode detector with a sensitive area of 0.28 mm² and a sensitive volume of 0.017 mm³, integrated into the dosimetric system at the CATANA proton therapy facility, routinely used for QA tests [21]. An in-house software allows the proton beam scanning along both the horizontal and vertical directions, with a step size ranging from 0.1 to 1 mm. The same software performs the calculation of the beam quality parameters. No bias voltage is applied.

–The Lynx (IBA Company, Germany) system, based on a gadolinium scintillating screen (0.4 mm thickness) coupled with a charge-coupled device (CCD). The scintillator active surface area is of 300 mm × 300 mm and an effective resolution of 0.5 mm in the X-Y plane is reached [22].

–The Timepix detector based on Medipix2 technology (CERN, Geneva, Switzerland) with 300 μm thick Si sensor. The sensor is divided into a number of 256 × 256, 55 μm wide square individual diodes/pixels [23,24].

3. Results and discussion

3.1. Inter-comparison at 62 MeV proton irradiation

The BP system was characterized comparing its performances as

respect the output of the imaging systems used in the quality assurance.

At CATANA results from BP were compared against the in-air, not polarized, silicon diode and the EBT3 radiochromic film, used as a reference in the clinical dosimetric practice. The normalized transversal dose profiles of the 62 MeV proton beam were extracted both in X (see Fig. 4) and Y directions. The Table 4 reports the estimation of the beam quality parameters described in Section 2.2.1, together with the experimental errors calculated as the standard deviation of their distributions around the central axes.

The profile size values, in terms of FWHM, are each other comparable within the experimental errors, with differences within 1%. The performances of the BP system result in quite good agreement with EBT3 films also in terms of field ratio (90%–50%), flatness and symmetry with relative differences up to 5%, while lateral penumbras (80%–20%) measured with the scintillator-based system are larger than those obtained with EBT3. This is most probably due mainly to scattering and/or diffusing phenomena inside the scintillator itself resulting in a light spread out and, therefore, into less sharp profile edges.

A more accurate estimation of the beam quality parameters with respect to the reference detector was obtained, in general, with the silicon diode. In this case, the relative differences in field size and lateral penumbras range between 10 and 30 μm , i.e. within the spatial resolution of the two detectors and they do not exceed 2.5%. Nevertheless, the real-time analysis of the BP system allows strong reduction of the acquisition and processing time (about 20 s against 5 and 30 min for the silicon diode and EBT3, respectively).

3.2. Inter-comparison at 100 MeV proton irradiation

In order to investigate the response of the system at higher energies, the BP tests were also performed with the 100 MeV proton beam at the Trento Proton Therapy facility. In this case, lateral profiles have been compared with those obtained with the Lynx detector, the Timepix and EBT3. Fig. 5 shows the comparison between their transversal dose profiles along the X direction. Because of the different tolerable fluences for the Timepix and the scintillator-based detector (of the order of 10^6 p/cm² and 10^{10} p/cm² respectively), slightly different beam currents were employed, without this leading to significant changes in the profiles shape as verified by means of EBT3 films.

Overall, all the field size values obtained with the different detectors are comparable with each other within the experimental errors, being the discrepancies with the reference detector below 3%. The FWHM value measured with the Timepix detector, in particular, agrees within 0.1% of discrepancy with the EBT3 radiochromic film, while the relative difference between the BP system and the EBT3 does not exceed 0.7%, as reported in detail in Table 5. The FWHM measured with the Lynx detector appears to be significantly larger than that obtained with the BP system, probably due to the higher Z of its scintillator material which originates a more diffusing light and/or to some smoothing-based algorithm implemented inside the acquisition system, resulting into a smearing and broadening of the beam lateral profile.

The lower spatial resolution of the Lynx detector appears evident from the comparison of its lateral penumbras as respect to the reference detector, being the relative differences significantly higher (more than 70%), while a discrepancy of less than 6% and 3.5% is recorded for the BP system and the Timepix respectively.

The flatness and the symmetry have been also estimated and reported in Table 5. In general, the Timepix detector shows a better accuracy in the transversal dose profile detection but a lower radiation hardness as compared to the BP system. The Lynx commercial detector is fast but its capability into reconstruct sharp dose profiles and estimate the beam quality parameters is less accurate from the spatial resolution point of view, at least for “small fields” investigated in this work, up to 30 mm $\times$ 30 mm. The EBT3 shows a good resolution (less than 0.2 mm) but it is not a real-time detector and more time-consuming for the analysis procedures.

The cross-check with the Timepix, the Lynx and EBT3 demonstrated the suitability of the BP system for the online transversal dose distribution characterization.

4. Conclusions

A real-time scintillator-based detector for daily QA tests in proton therapy has been developed at the LNS of INFN and characterized with proton beams up to 100 MeV energy.

The device shows promising qualities as control system for 2-D relative dosimetry thanks to its linear response in a wide dose rate range, its stability into reconstruct sharp dose profiles and its short acquisition and processing time. A validation with respect to EBT3 radiochromic films, used as a reference, and a comparison with other devices have also been performed. For the investigated energy of 62 MeV, the relative differences between the beam profiling system and the EBT3 don't exceed 6% for FWHM, 90/50 ratio, flatness and symmetry, while lateral penumbras measured with the scintillator are systematically larger, with deviations up to about 25%. At higher energy, 100 MeV, the agreement seems to be better thanks to the lower light yield, being the differences for all the investigated parameters within 6%. Lower and significantly higher discrepancies show the Timepix and the Lynx respectively, confirming, as expected, that silicon detectors could be preferred to the scintillation ones if extremely low beam currents are employed.

Although this study was focused on the characterization of the beam profiling system with proton beams, the use of the device can be obviously extended to other ion beams, where the lateral penumbra of the beam is expected to be sharper. Further measurements to investigate the detector performances at higher proton beam energies (up to 200 MeV) are currently ongoing, together with the optimization of the setup configuration in terms of reduction of scattering and/or light diffusion throughout the scintillator itself and attenuation of photons reflection inside the box, in order to reduce the light spread out and make the BP system ready for beam commissioning measurements. Despite of several hundreds of kilo grays have been seen by the screen, no aging effect of the scintillator are observed so far, even though a systematic radiation damage study has not carried out yet. A “large-field” system for broader beams, up to 300 mm $\times$ 300 mm sensitive region, useful for any kind of cancer and clinical condition or different applications, is also currently under design and test. At the same time, a study aimed at reducing the size of the system is ongoing as well.

In summary, the reported results show that the beam profiling system developed at the INFN-LNS is potentially suitable for application in QA measurements as well as for quick and real-time centering and optimization of the proton beam by the cyclotron console operators before patient treatments. Specifically, experimental results demonstrate that the obtained accuracy is very close to the silicon pixelated detector (Timepix) and to existing dosimetric devices (silicon diode and EBT3 films); finally, a significantly higher resolution was observed with respect to the widely used Lynx detector. A comparison with a small ionization chamber and a validation with the results from Monte-Carlo simulations will be carried out in our future studies.

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